

# Redesigning the Bad Graph

2023-03-03

## R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com> (<http://rmarkdown.rstudio.com>).

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Redesigning of the bad graph: This is link for the bad graph: <https://money.cnn.com/interactive/news/handgun-homicides/> (<https://money.cnn.com/interactive/news/handgun-homicides/>)

We would like to redesign the above pie charts into an interactive plot, by separating the regions with in USA map

```
library(tidyverse)
```

```
## — Attaching packages ————— tidyverse 1.3.2 —
## ✓ ggplot2 3.4.1      ✓ purrr   1.0.1
## ✓ tibble  3.1.8      ✓ dplyr   1.1.0
## ✓ tidyr   1.3.0      ✓ stringr 1.5.0
## ✓ readr   2.1.3      ✓ forcats 1.0.0
## — Conflicts ————— tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
```

```
library(ggplot2)
library(dplyr)
library(plotly)
```

```
##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##   last_plot
##
## The following object is masked from 'package:stats':
##
##   filter
##
## The following object is masked from 'package:graphics':
##
##   layout
```

```
library(maps)
```

```
##  
## Attaching package: 'maps'  
##  
## The following object is masked from 'package:purrr':  
##  
##      map
```

```
library(usmap)
```

```
# Data extracted from dot plot
```

```
edu_cost <- c(8500,8000,9000,9200,8800,12000,10500,8700,9100,  
             7500,10200,9200,9300,9300,9000,9700,11500,12500,9700,  
             9700,9500,9800,9900,8500,11500,16300,18500,8000,  
             9850,9800,6500,12500,13500,7500,14000,9600,8500,10200,10000)  
prison_cost <- c(16500,23500,23000,47000,29000,48000,32000,19000,19800,  
                17000,36000,13500,31500,15500,13000,15000,45000,36000,  
                27000,40000,21500,29000,34000,19000,32100,54000,58700,  
                28500,38600,23500,17500,41300,48500,20000,48700,24600,  
                45000,25000,37600)  
differ <- c(8000, 15500,14000,37800,20200,36000,21500,10300,10700,9500,25800,4300,22200,  
           6200,4000,5300,33500,23500,17300,30300,12000,19200,24100,10500,20600,37700,  
           40200,20500,28750,13700,11000,28800,35000,12500,34700,15000,36500,14800,27600)  
states <- c("Alabama", "Arizona","Arkansas","California","Colorado","Connecticut",  
            "Delware","Florida","Georgia","Idaho","Illinois","Indiana","Iowa","Kansas",  
            "Kentucky","Louisiana","Maine","Maryland","Michigan","Minnesota","Missouri",  
            "Montana","Nebraska","Nevada","New Hampshire","New Jersey","New York","North Carolina",  
            "North Dokota","Ohio","Oklahoma","Pennsylvania","Rhode Island","Texas","Vermont","Virginia",  
            "Washington","West Virginia","Wisconsin")  
  
dataset1 <- cbind(edu_cost,prison_cost)  
dataset1
```

| ##       | edu_cost | prison_cost |
|----------|----------|-------------|
| ## [1,]  | 8500     | 16500       |
| ## [2,]  | 8000     | 23500       |
| ## [3,]  | 9000     | 23000       |
| ## [4,]  | 9200     | 47000       |
| ## [5,]  | 8800     | 29000       |
| ## [6,]  | 12000    | 48000       |
| ## [7,]  | 10500    | 32000       |
| ## [8,]  | 8700     | 19000       |
| ## [9,]  | 9100     | 19800       |
| ## [10,] | 7500     | 17000       |
| ## [11,] | 10200    | 36000       |
| ## [12,] | 9200     | 13500       |
| ## [13,] | 9300     | 31500       |
| ## [14,] | 9300     | 15500       |
| ## [15,] | 9000     | 13000       |
| ## [16,] | 9700     | 15000       |
| ## [17,] | 11500    | 45000       |
| ## [18,] | 12500    | 36000       |
| ## [19,] | 9700     | 27000       |
| ## [20,] | 9700     | 40000       |
| ## [21,] | 9500     | 21500       |
| ## [22,] | 9800     | 29000       |
| ## [23,] | 9900     | 34000       |
| ## [24,] | 8500     | 19000       |
| ## [25,] | 11500    | 32100       |
| ## [26,] | 16300    | 54000       |
| ## [27,] | 18500    | 58700       |
| ## [28,] | 8000     | 28500       |
| ## [29,] | 9850     | 38600       |
| ## [30,] | 9800     | 23500       |
| ## [31,] | 6500     | 17500       |
| ## [32,] | 12500    | 41300       |
| ## [33,] | 13500    | 48500       |
| ## [34,] | 7500     | 20000       |
| ## [35,] | 14000    | 48700       |
| ## [36,] | 9600     | 24600       |
| ## [37,] | 8500     | 45000       |
| ## [38,] | 10200    | 25000       |
| ## [39,] | 10000    | 37600       |

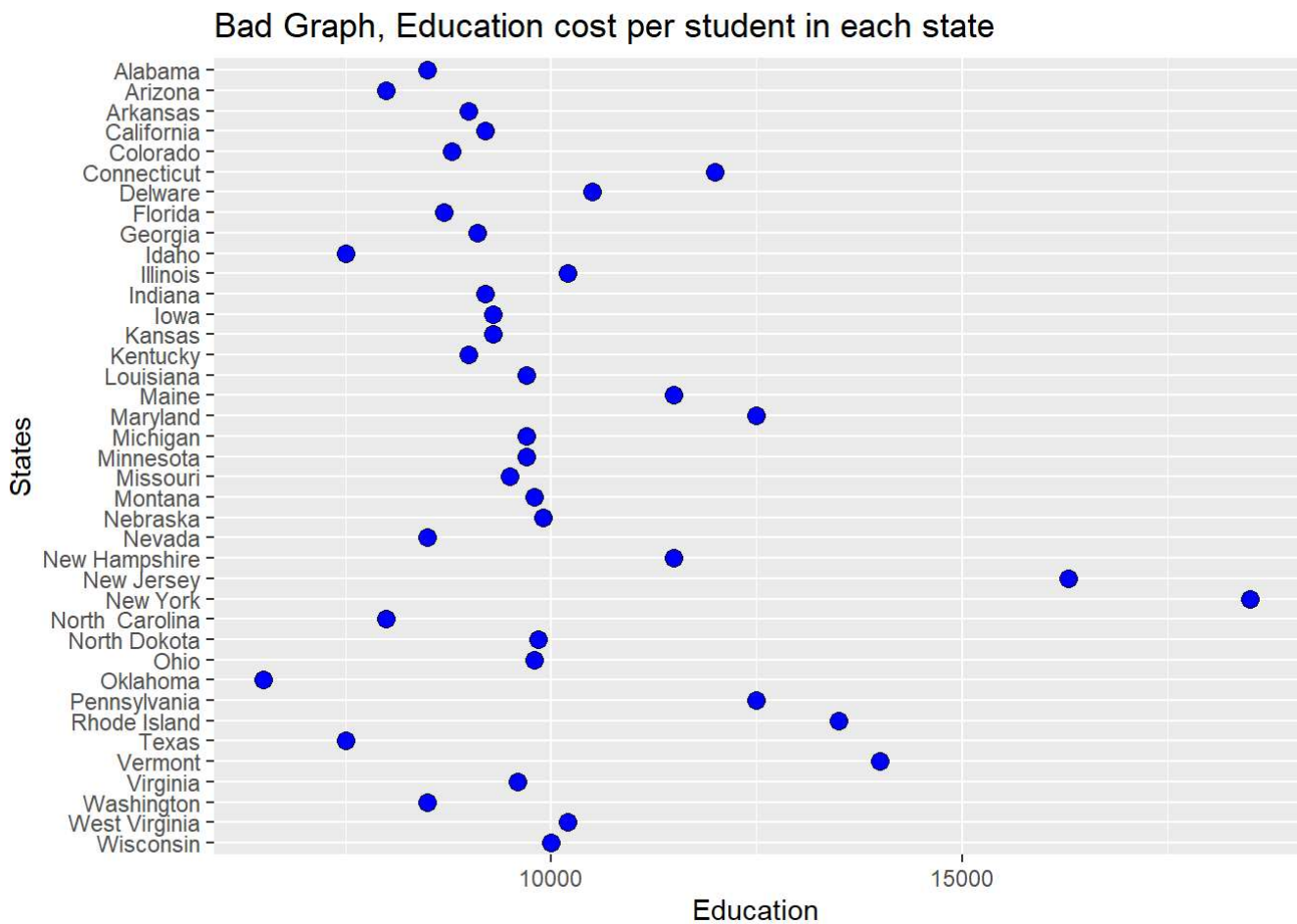
```
names<- c("Education Cost", "Prison Cost")
```

```
colnames(dataset1) <- names
rownames(dataset1) <- states
```

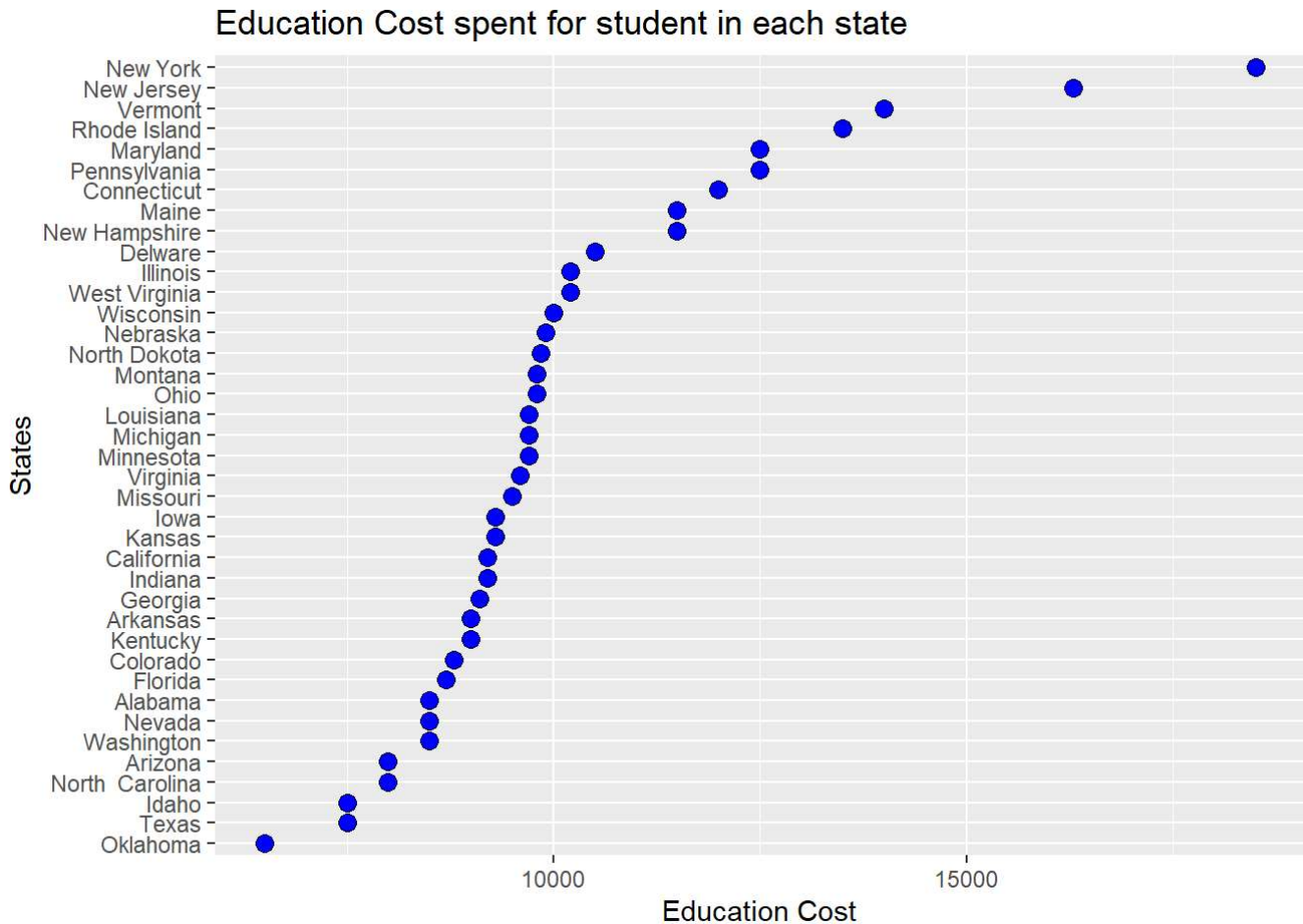
```
dataset2 <- as.data.frame(dataset1) # make a data.frame
types = rownames(dataset2)
dataset2$States = factor(states, ordered = T, levels = rev(states))
```

```
rownames(dataset2) = NULL # removing row names from the data set (Optional)
```

```
dataset2$difference <- dataset2$`Prison Cost` - dataset2$`Education Cost`
ggplot(dataset2, aes(x= edu_cost, y= States)) + geom_point(shape = 21,
  fill = "blue",
  color = "black",
  size = 3) + labs(x = "Education" , title = "Bad Graph, Education cost per student in each st
ate")
```

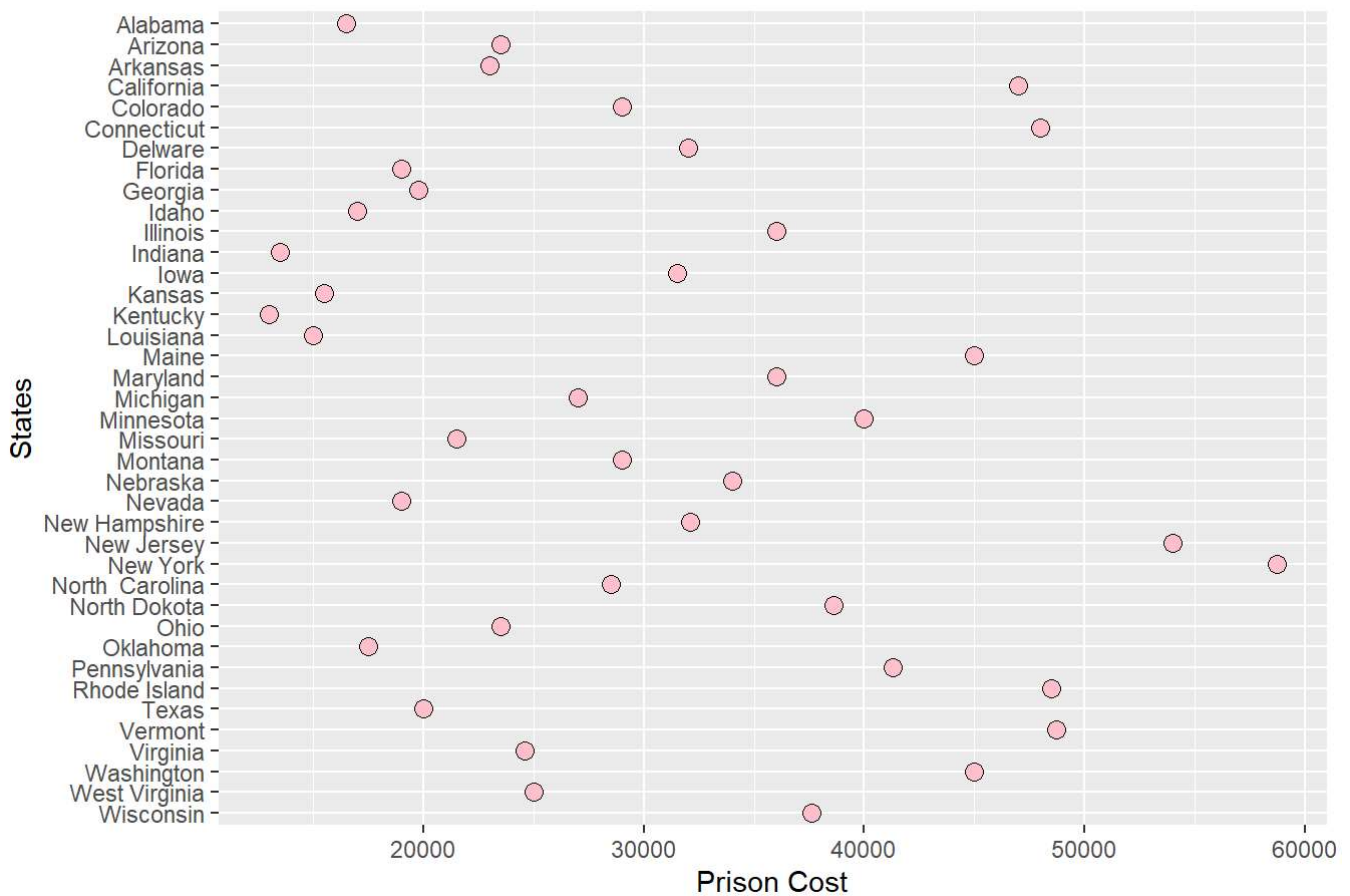


```
student_data <- ggplot(dataset2, aes(x = edu_cost, y = reorder(States, edu_cost))) +
  geom_point(
    shape = 21,
    fill = "blue",
    size = 3,
    color = "black"
  ) +
  labs(x = "Education Cost",
    y = "States",
    title = "Education Cost spent for student in each state")
student_data
```



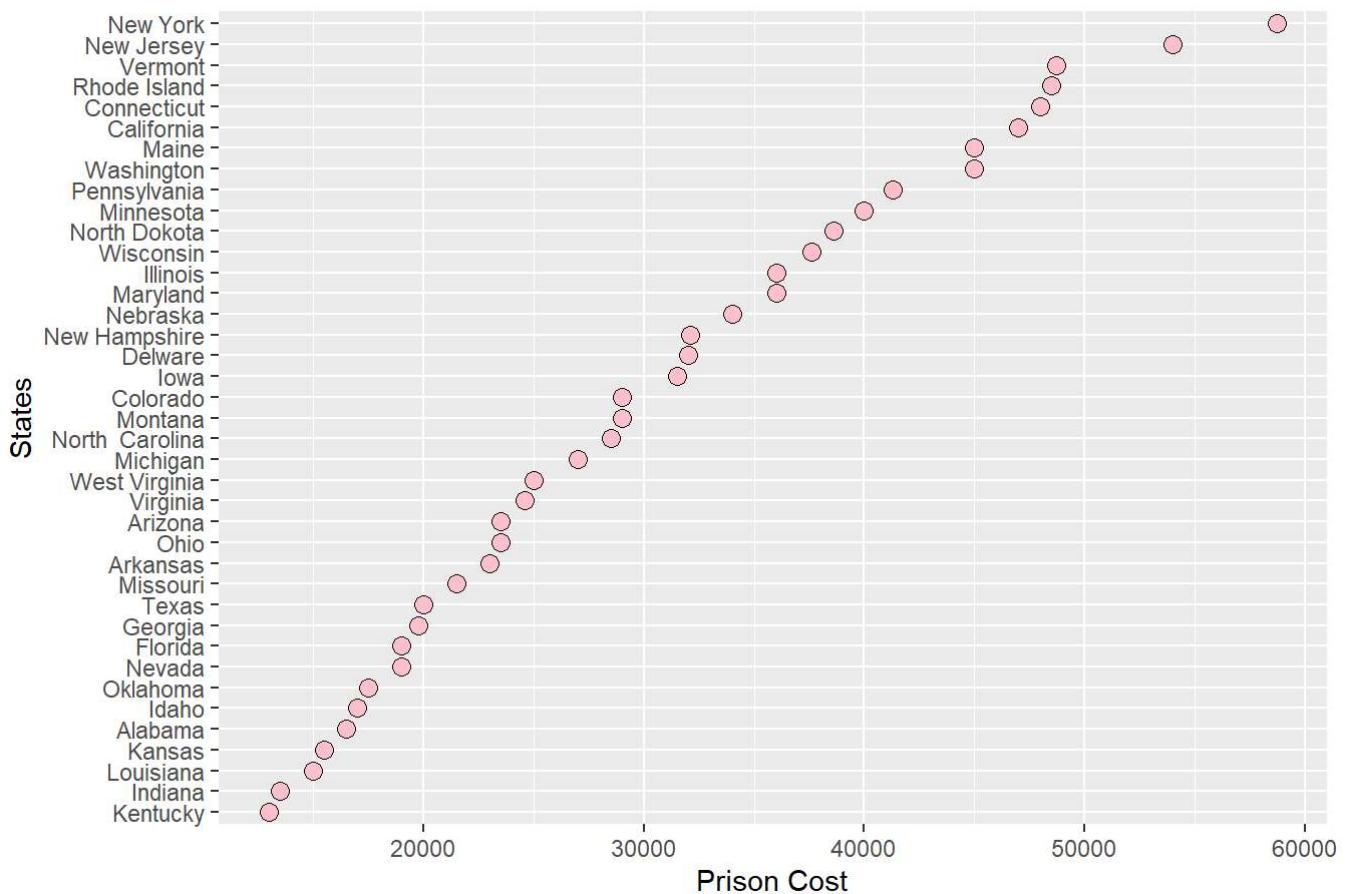
```
ggplot(dataset2, aes(x = prison_cost, y = States)) +
  geom_point(
    shape = 21,
    fill = "pink",
    size = 3,
    color = "black"
  ) +
  labs(x = "Prison Cost",
    y = "States",
    title = "Bad Graph, Prison Cost spent for student in each state")
```

Bad Graph, Prison Cost spent for student in each state



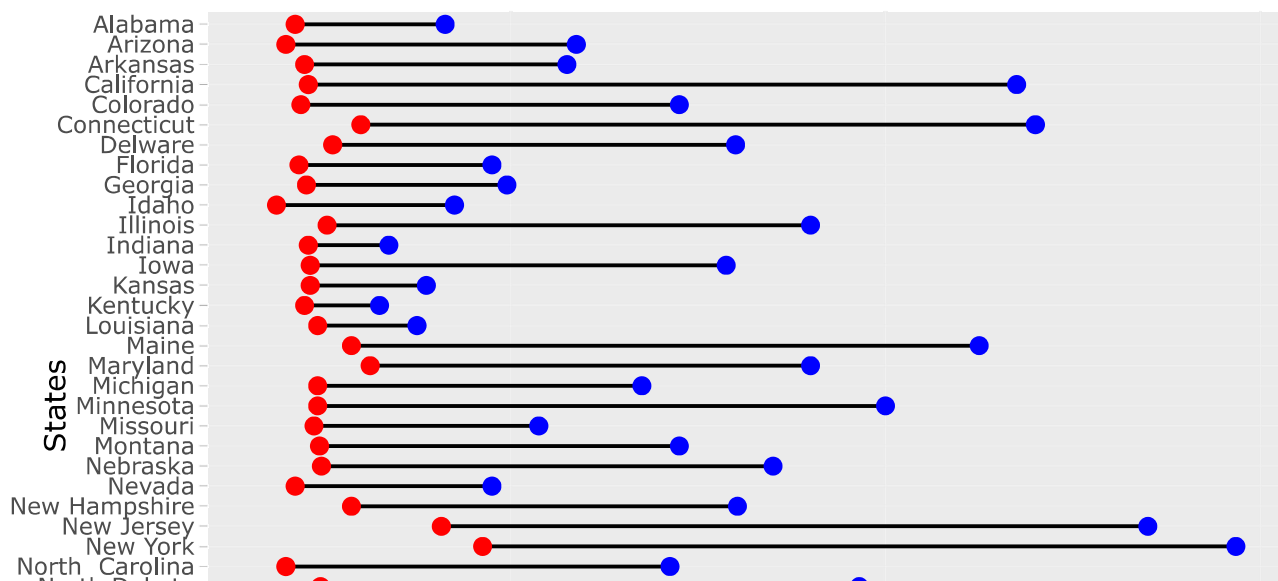
```
prisoner_data <- ggplot(dataset2, aes(x = prison_cost, y = reorder(States, prison_cost))) +
  geom_point(
    shape = 21,
    fill = "pink",
    size = 3,
    color = "black"
  ) +
  labs(x = "Prison Cost",
       y = "States",
       title = "Prison Cost spent for student in each state")
prisoner_data
```

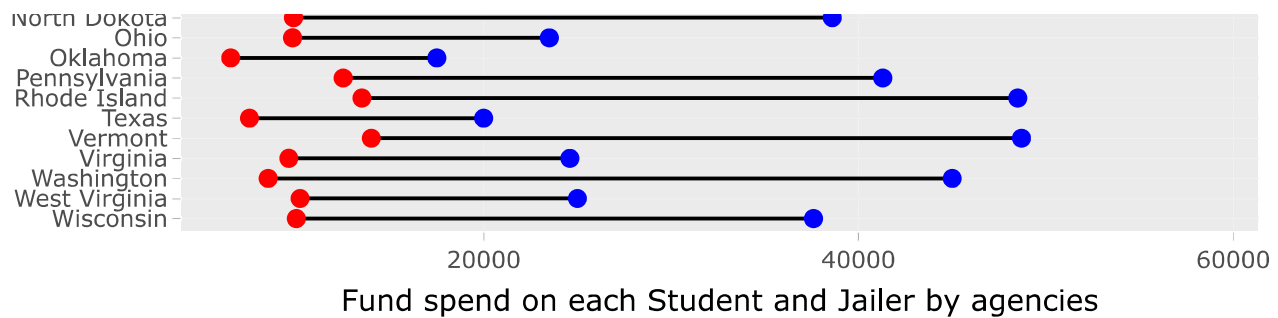
Prison Cost spent for student in each state



```
p <- ggplot(dataset2, aes(y= States, x= edu_cost, xend = prison_cost)) +
  geom_segment(aes(x= edu_cost, xend = prison_cost, yend = States)) +
  geom_point(aes(x= edu_cost), size = 2 , color = "red") +
  geom_point(aes(x= prison_cost), size = 2, color = "blue") +
  labs(title = "Amount spent by each state on Student and Prisoner", x="Fund spend on each Student and Jailer by agencies")
ggplotly(p)
```

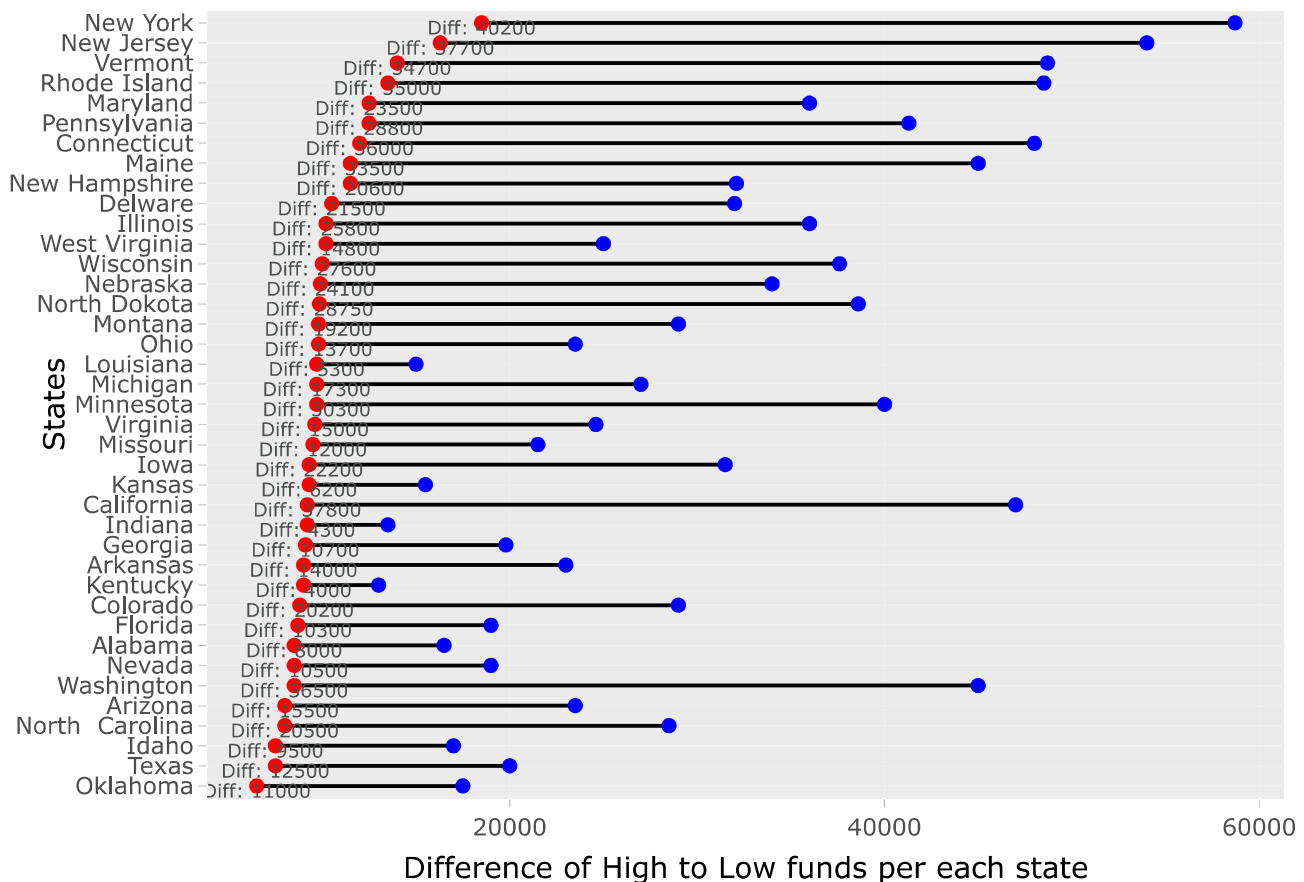
Amount spent by each state on Student and Prisoner





```
p_diff <-ggplot(dataset2, aes(y= reorder(States, edu_cost), x= edu_cost, xend = prison_cost)) +
  geom_segment(aes(x= edu_cost, xend = prison_cost, yend = States)) +
  geom_point(aes(x= edu_cost), size = 1.5 , color = "red") +
  geom_point(aes(x= prison_cost), size = 1.5, color = "blue") +
  geom_text(data = dataset2, aes(label = paste("Diff:",differ), x= edu_cost, y= States),
    color = "#4a4e4d",
    size = 2.5,
    nudge_y = -0.35,
    family = "Segoe UI Semibold")+
  labs(title = "Difference of Amount spent by each state on Student and Prisoner", x=
"Difference of High to Low funds per each state", y= "States")
ggplotly(p_diff)
```

## Difference of Amount spent by each state on Student and Prisoner





```

us_states = map_data("state");

#map each state to towards there directions
us_states$direction[us_states$region %in%
  c("connecticut", "maine", "massachusetts", "new hampshire", "rhode island", "vermont",
    "new jersey", "new york", "pennsylvania")] <- "East Coast"
us_states$direction[us_states$region %in%
  c("illinois", "indiana", "michigan", "ohio", "wisconsin", "iowa", "kansas", "minnesot
a", "missouri", "nebraska", "north dakota", "south dakota")] <- "Midwest"
us_states$direction[us_states$region %in%
  c("delaware", "florida", "georgia", "maryland", "north carolina", "south carolina", "v
irginia", "district of columbia", "west virginia", "alabama", "kentucky", "mississippi", "tennes
see", "arkansas", "louisiana", "oklahoma", "texas")] <- "Gulf Coast"
us_states$direction[us_states$region %in%
  c("alaska", "california", "hawaii", "oregon", "washington", "arizona", "colorado", "id
aho", "montana", "nevada", "new mexico", "utah", "wyoming")] <- "West Coast"

#data extracted from the pie charts

states <- c("alabama", "alaska", "arizona", "arkansas", "california", "colorado", "connecticut",
  "delware", "florida", "georgia", "hawaii", "idaho", "illinois", "indiana", "iowa", "kansa
s",
  "kentucky", "louisiana", "maine", "maryland", "massachusetts", "michigan", "minnesot
a", "mississippi", "missouri",
  "montana", "nebraska", "nevada", "new hampshire", "new jersey", "new mexico", "new yor
k", "north carolina",
  "north dokota", "ohio", "Oklahoma", "oregon", "pennsylvania", "rhode island", "south ca
rolina", "south dakota", "tennessee", "texas", "utah", "vermont", "virginia", "washington", "west virgi
nia", "wisconsin", "wyoming", "district of columbia")

percofdeaths <- c(0,31,74,47,71,53,57,64,0,88,0,88,97,63,37,42,77,93,25,96,43,59,84,88,57,29,83,
61,17,88,75,89,70,50,54,76,33,81,20,57,60,70,71,58,50,53,73,53,75,64,48)

deathscount <- c(0,5,165,52,866,39,54,18,0,326,0,15,364,115,7,31,77,372,3,262,52,267,36,121,158,
2,35,46,1,238,45,394,235,3,187,99,13,379,1,126,3,172,497,15,2,11,58,23,60,7,43)

prisonercost <- c(17000, 52633,22000,21900,64642,39303,62159,39080,19900,20000,0,18000,36000,140
00,37908,17000,13000,16000,45000,44601,55170,27000,41366,0,21500,28000,34000,19000,33000,61603,3
6832,69355,28000,37000,25000,17000,44021,42727,58564,0,0,0,20500,28000,57615,24000,37841,25000,3
8644,0,0)

dataone <- cbind(percofdeaths,deathscount,prisonercost)

names<- c("perofdeaths","deathscount","prisonercost")

colnames(dataone) <- names
rownames(dataone) <- states

datatwo <- as.data.frame(dataone) # make a data.frame

```

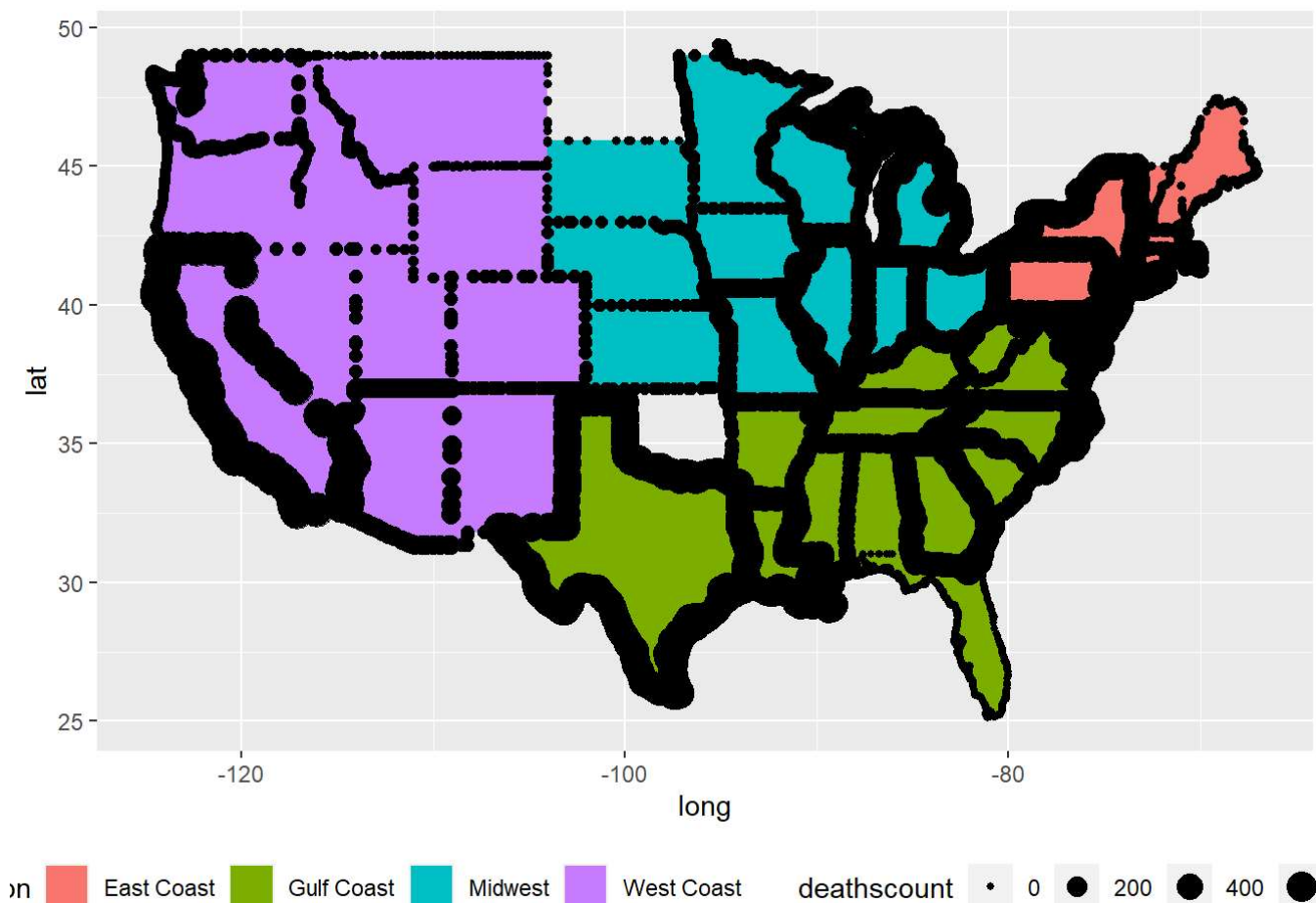
```

types = rownames(datatwo)
datatwo$region = factor(states, ordered = T)
datatwo$fips<-c("AL","AZ","AR","AK","CA","CO","CT","DE","FA","GA","HI","ID","IL","IN","IA","K
S","KY","LA","ME","MD","MA","MI","MN","MS","MO","MT","NE","NV","NH","NJ","NM","NY","NC","ND","O
H","OK","OR","PA","RI","SC","SD","TN","TX","UT","VT","VA","WA","WV","WI","WY","DC")

mapdata<-inner_join(us_states, datatwo, by = "region")

map1 <- ggplot()
map1 = map1 + geom_polygon(data=mapdata, aes(x=long, y=lat, group=group, fill=direction))+ geom_
point(data= mapdata, aes(x= long, y= lat, size = deathscout))+theme(legend.position = "bottom")
man1

```



```

mapdata$lat.x[mapdata$direction=="East Coast"] <- mapdata$lat[mapdata$direction == "East Coast"]
mapdata$long.x[mapdata$direction=="East Coast"] <- mapdata$long[mapdata$direction == "East Coast"] +5

mapdata$lat.x[mapdata$direction=="Midwest"] <- mapdata$lat[mapdata$direction == "Midwest"]
mapdata$long.x[mapdata$direction=="Midwest"] <- mapdata$long[mapdata$direction == "Midwest"]

mapdata$lat.x[mapdata$direction=="Gulf Coast"] <- mapdata$lat[mapdata$direction == "Gulf Coast"] -3
mapdata$long.x[mapdata$direction=="Gulf Coast"] <- mapdata$long[mapdata$direction == "Gulf Coast"]

mapdata$lat.x[mapdata$direction=="West Coast"] <- mapdata$lat[mapdata$direction == "West Coast"] -2
mapdata$long.x[mapdata$direction=="West Coast"] <- mapdata$long[mapdata$direction == "West Coast"] -10

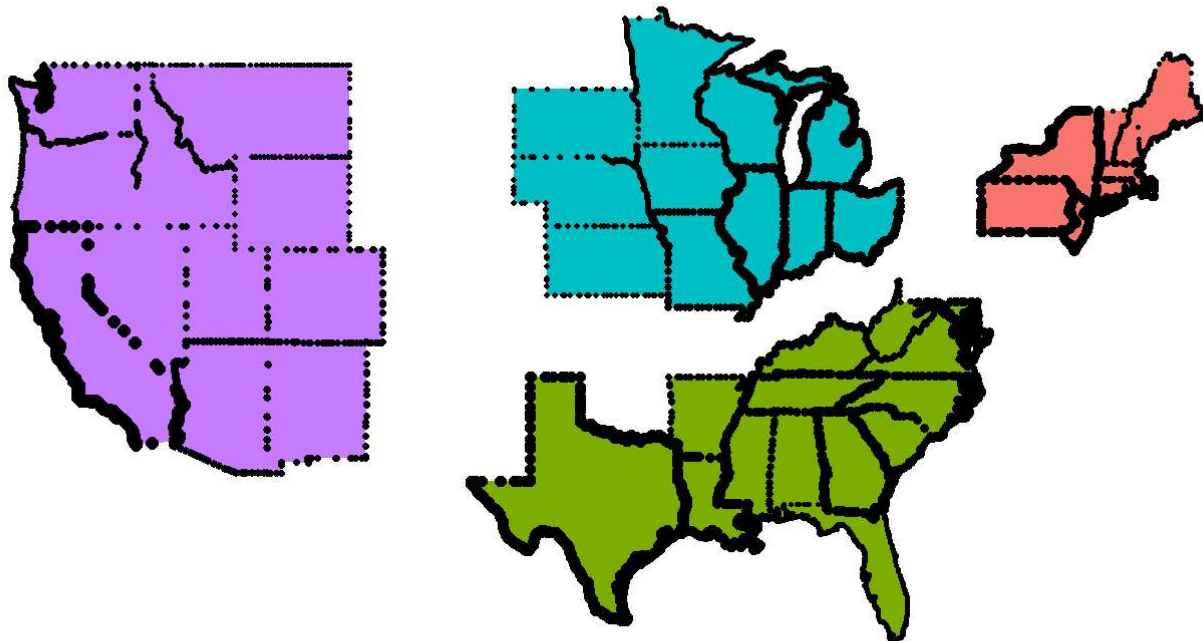

Map1.1<- ggplot(mapdata, aes(x=long.x, y=lat.x)) +
  geom_polygon(aes(group = group, fill=direction))+
  geom_point(data= mapdata, aes(x= long.x, y= lat.x, size = deathscout))+scale_size(name = " ",
range=c(0.5,2))+guides(size=guide_legend("deaths"))+theme(legend.position="bottom")+
  labs(title="United States region wise demographics",
        subtitle='Handgun Deaths',
        x="longitude",y="latitude",
        caption=c('Source: 2011 F.B.I. Crime in the United States report'))+theme(panel.background = element_blank(),
  panel.grid = element_blank(),
  axis.line = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank(),
  axis.text = element_blank())

plot(Map1.1)

```

## United States region wise demographics

### Handgun Deaths



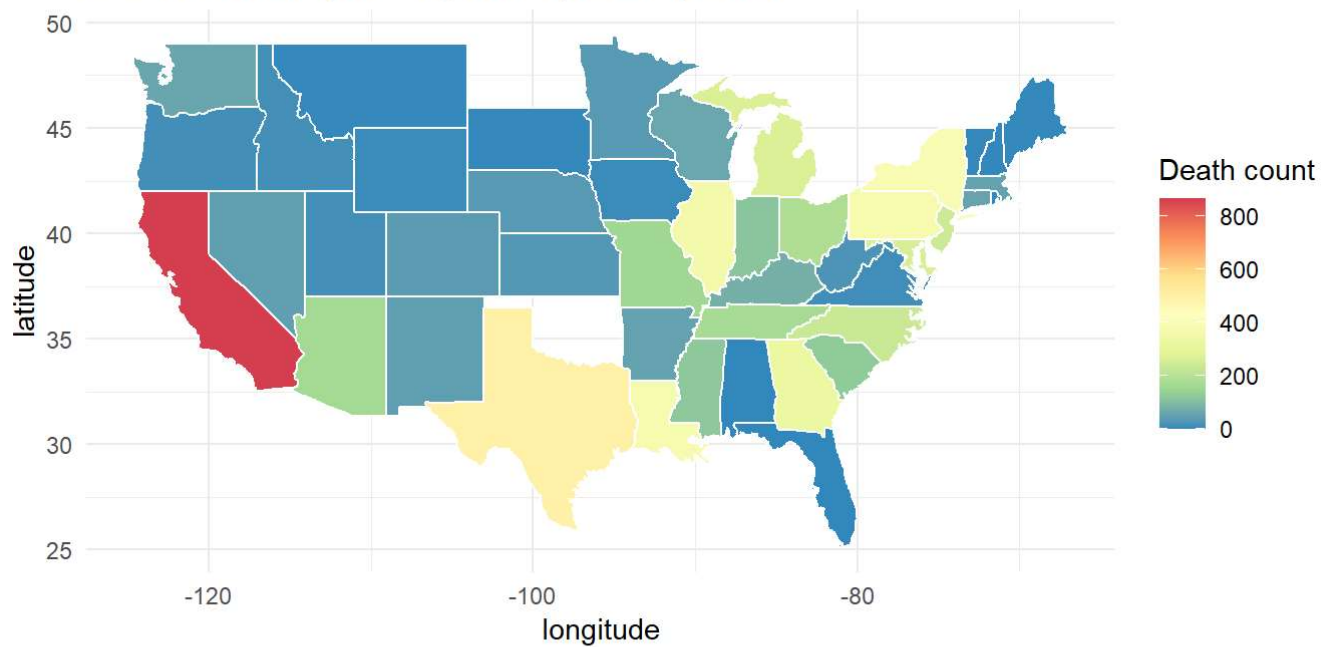
• 0 • 200 • 400 • 600 • 800 direction East Coast Gulf Coast Midwest W

Source: 2011 F.B.I. Crime in the United States report

```
usa <- map_data("usa")
states <- map_data("state")

ggplot(mapdata, aes(map_id = region)) + geom_map(aes(fill = deathcount ), map = states, col =
"white") +
  expand_limits(x = states$long, y = states$lat) +
  scale_fill_distiller(name = "Death count", palette = "Spectral") +
  labs(title = "US Death Rate per 100,000 Population, 2011", x = "longitude", y = "latitude",
        caption = "Source: 2011 F.B.I. Crime in the United States report") +
  coord_fixed(1.3) +
  theme_minimal()
```

## US Death Rate per 100,000 Population, 2011



Source: 2011 F.B.I. Crime in the United States report

```
westcoast<-plot_usmap(include = c("AL", "CA", "HI", "AZ", "OR","WA","MN","ID","CO","NM", "WY","U
T","NV")) +
  labs(title = "West Coast Region") +
  theme(panel.background = element_rect(color = "blue"))
plot(westcoast)
```

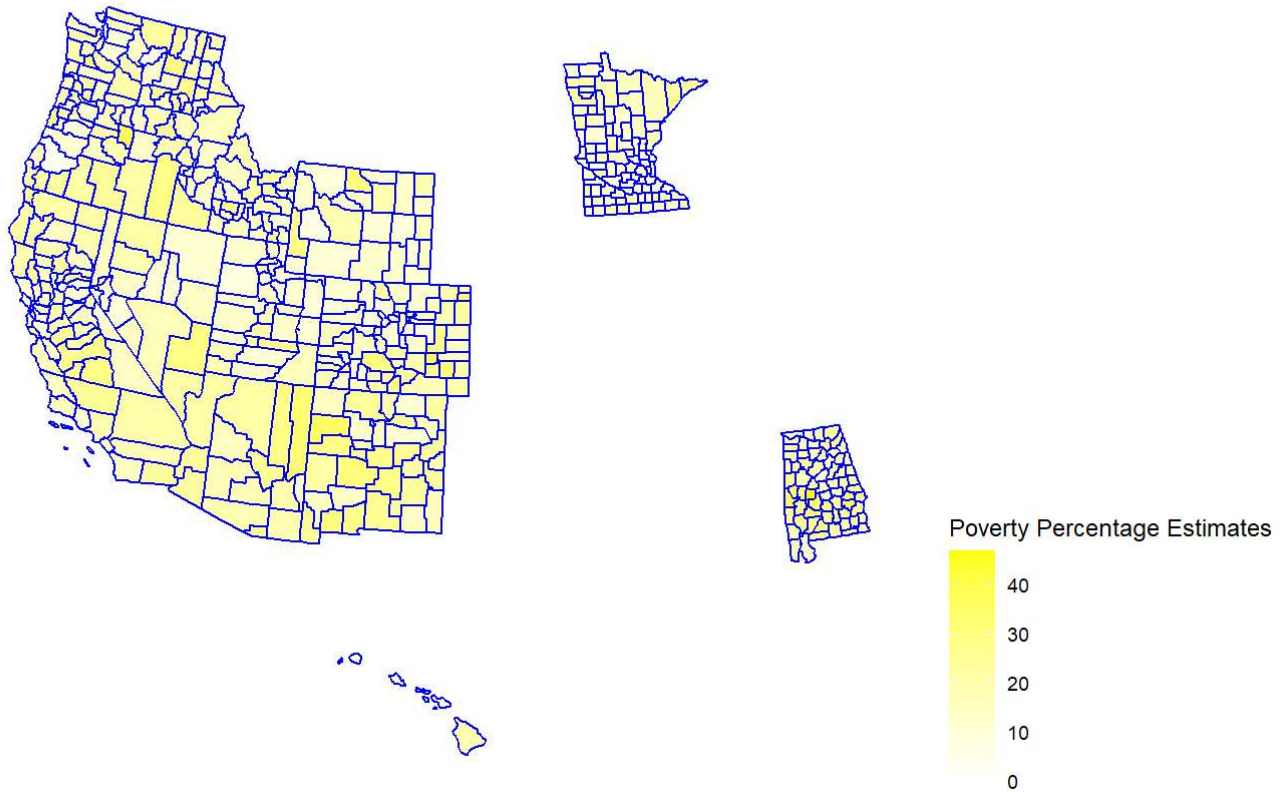
## West Coast Region



```
plot_usmap(data = countypov, values = "pct_pov_2014", include = c("AL", "CA", "HI", "AZ", "OR", "WA", "MN", "ID", "CO", "NM", "WY", "UT", "NV"), color = "blue") +  
  scale_fill_continuous(low = "white", high = "yellow", name = "Poverty Percentage Estimates", labels = scales::comma) +  
  labs(title = "New England Region", subtitle = "Poverty Percentage Estimates for New England Counties in 2011") +  
  theme(legend.position = "right")
```

## New England Region

Poverty Percentage Estimates for New England Counties in 2011



```
library(viridis)
```

```
## Loading required package: viridisLite
```

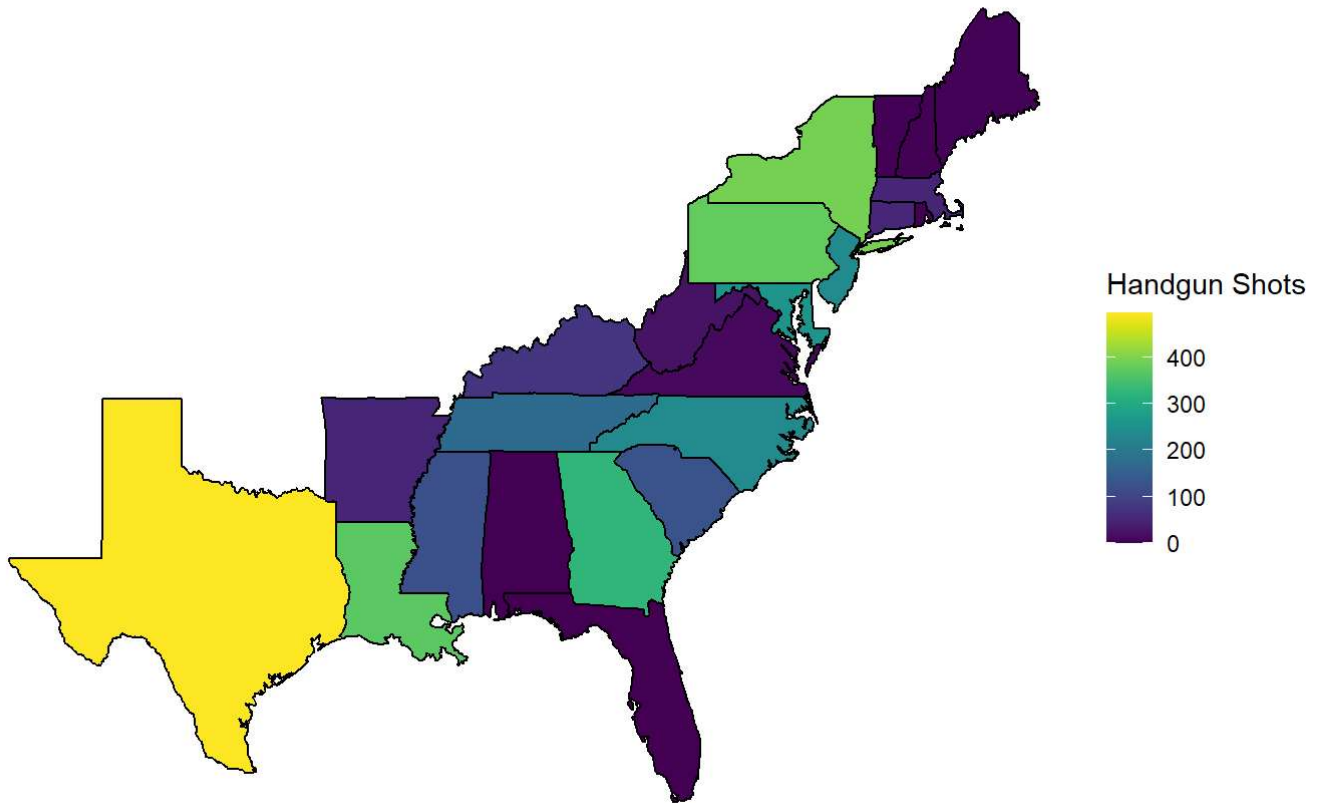
```
##  
## Attaching package: 'viridis'
```

```
## The following object is masked from 'package:maps':  
##  
## unemp
```

```
Coastalregion <- mapdata %>%  
  filter(direction %in% c("East Coast", "Gulf Coast"))  
  
TX_to_NY=ggplot(Coastalregion,aes(x = long, y = lat, group = group, fill = deathcount)) +  
  geom_polygon(color = "black")+  
  theme_void()+scale_fill_viridis(name = "Handgun Shots")+labs(title= "combined count for Gulf and East coasts", subtitle = "Handgun shots per each state")  
  
plot(TX_to_NY)
```

combined count for Gulf and East coasts

Handgun shots per each state

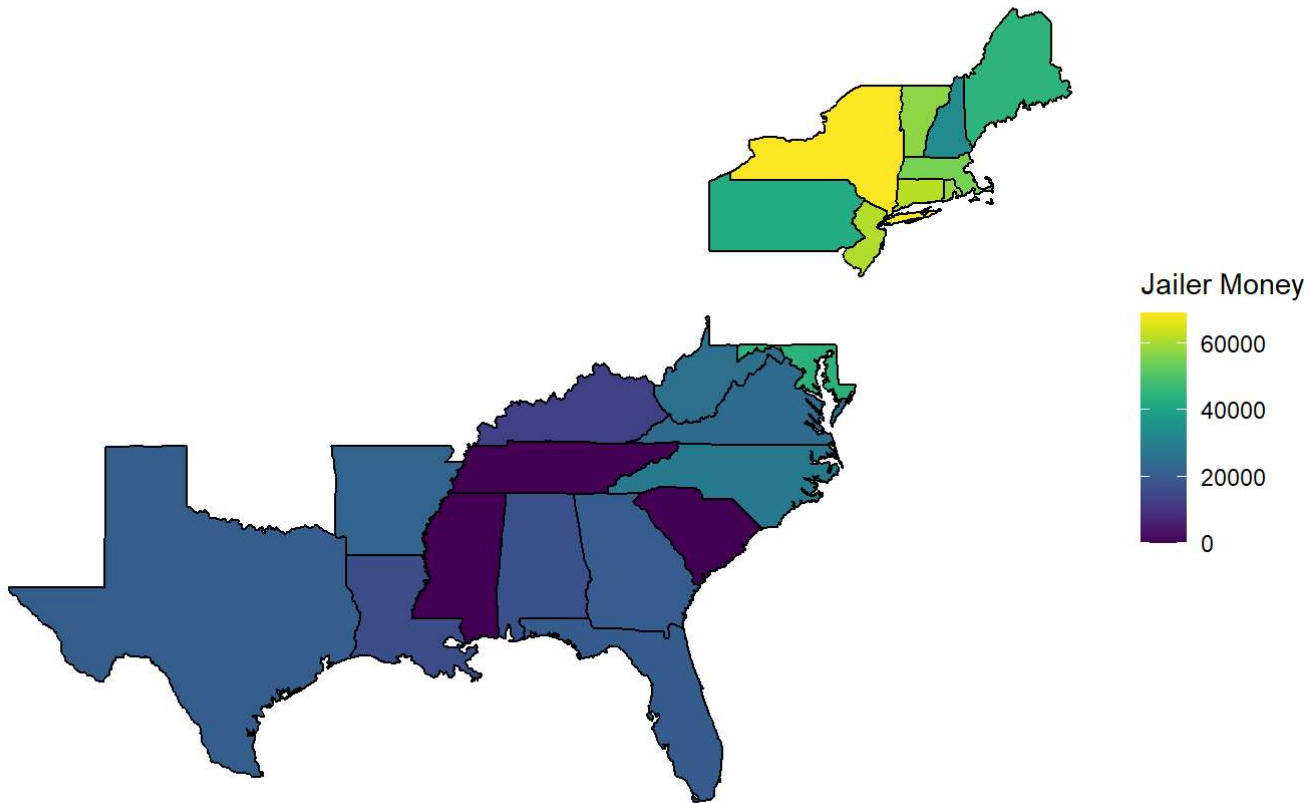


```
TX_and_NY=ggplot(Coastalregion, aes(x = long, y = lat.x, group = group, fill = prisonercost)) +  
  geom_polygon(color = "black")+theme_void()+scale_fill_viridis(name = "Jailer Money")+labs(titl  
e= "combined count for Gulf and East coasts", subtitle ="USA spent on jailers in each state")  
plot(TX_and_NY)
```



combined count for Gulf and East coasts

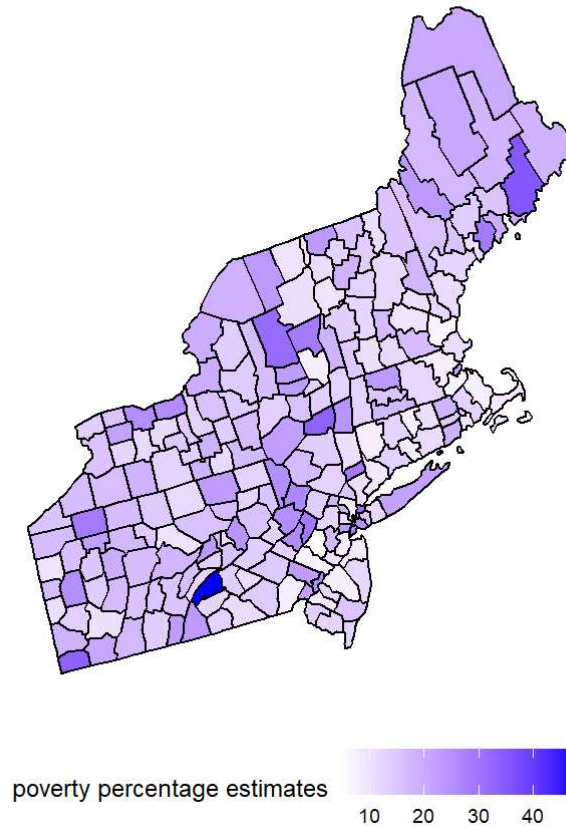
USA spent on jailers in each state



```
plot_usmap( data = countypov, values = "pct_pov_2014", include = c("CT", "ME", "MA", "NH", "RI", "VT", "NJ", "NY", "PA"), color = "black")+ scale_fill_continuous(low = "white", high = "blue", name = "poverty percentage estimates", label = scales::comma) + labs(title = "East Coast", subtitle = "Poverty in Coastal region in 2011") + theme(legend.position = "bottom")
```

## East Coast

Poverty in Coastal region in 2011



```
plot_usmap( data = countypov, values = "pct_pov_2014", include = c("TX","AL","FA","LA","MS","D  
E", "GA", "MD", "NC", "SC", "VA", "DC", "WV", "AL", "KY", "TN", "AK", "OK"),color = "black") +  
  scale_fill_continuous(low = "white", high = "red", name = "poverty percentage estimates", labe  
l = scales::comma) +  
  labs(title = "Gulf Coast", subtitle = "Poverty in Coastal region in 2011") +  
  theme(legend.position = "bottom")
```

## Gulf Coast

Poverty in Coastal region in 2011

